

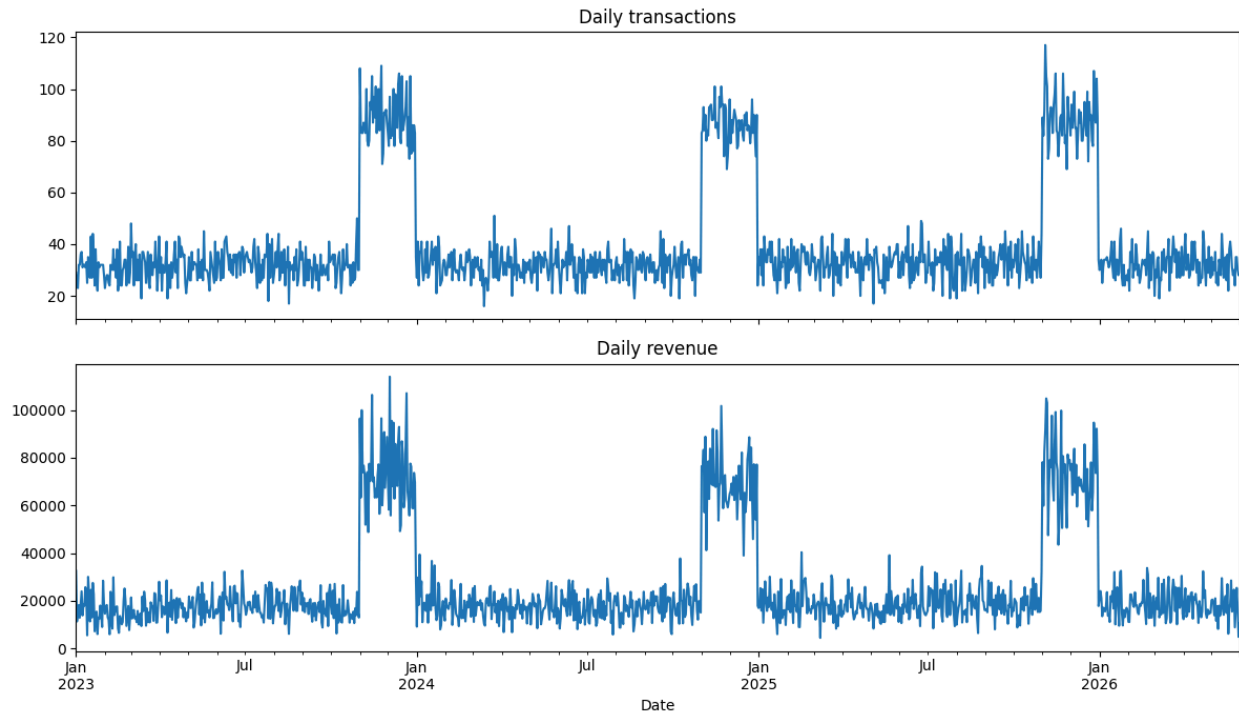
E-commerce Demand Forecasting: Quantifying the Nov–Dec Peak

Daily demand analysis on 50,000 global e-commerce transactions (Jan 2023 – May 2026). The project measures how large the Nov 1 – Dec 30 peak is, tests whether the peak window and the statistical model are justified, benchmarks six forecasting approaches with leakage-free validation, and turns the results into planning numbers for a peak season.

Main finding: peak season is a single regime shift. Orders rise 2.75× and revenue rises 4.03×, and a simple peak/normal regime mean forecasts daily revenue close to the best possible regime-based forecast, and no worse than LightGBM within statistical noise.

Headline results

Question	Result
How much does volume rise in peak?	2.75× daily transactions (95% CI 2.70–2.80)
How much does revenue rise?	4.03× daily revenue = 2.75× more orders × 1.46× higher order value
Is the Nov 1 – Dec 30 window justified?	Yes. It is the best-fitting window in a scan run on pre-2024-11-01 data only
Is a Poisson model appropriate?	Yes. Dispersion 0.97; Cameron–Trivedi p = 0.16
Best forecaster?	Regime mean (peak/normal). 29–39% lower MAE than seasonal naive
Does LightGBM beat it?	No. Statistically tied in both peak folds, and worse in the normal-period fold
Peak-season revenue baseline	4.33M per 60 days (past seasons: 4.17M–4.42M)



Data

- 50,000 transactions, 15 columns, 2023-01-01 to 2026-06-01 (the final date is excluded from the analysis)
 - Validation: 0 missing values, 0 duplicate transaction IDs, $\text{Unit_Price} \times \text{Quantity} \times (1 - \text{Discount}) = \text{Total_Revenue}$ holds for 100% of rows
 - Aggregated to a daily series: transactions, revenue, units, average discount, average order value (AOV)
 - **Peak** = Nov 1 – Dec 30 each year, defined once as a constant in the notebook
-

Method

1. **Validate the data**, then aggregate to daily level.
2. **Detect regime changes.** Binary segmentation on daily transactions finds changepoints at 2024-11-01, 2024-12-31, 2025-11-01 and 2025-12-31 (plus two near the 2023 season boundaries), which line up with the assumed peak window.
3. **Validate the peak window without leakage.** A grid of start and end dates was scored with a weekday-adjusted Poisson GLM using only data before 2024-11-01, so no test period influenced the choice.

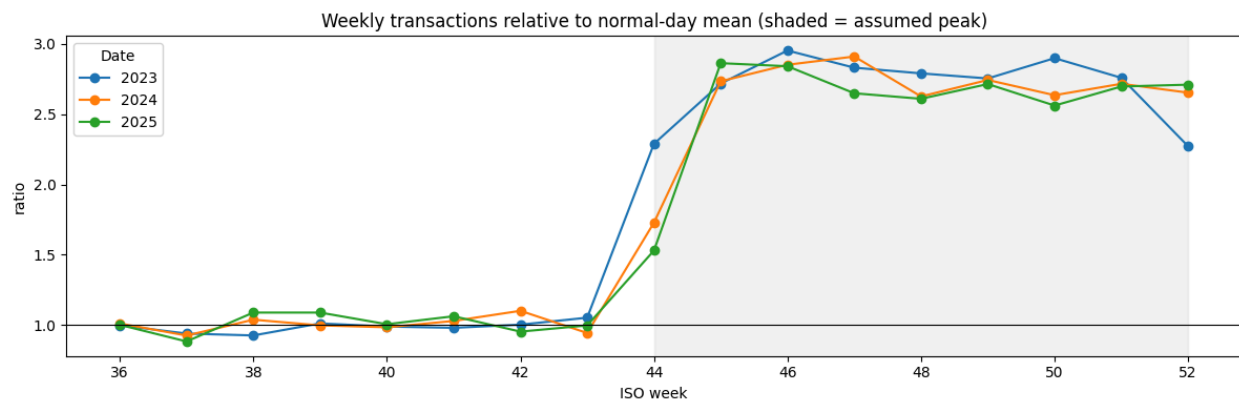
4. **Check the model assumptions.** Overdispersion test, robust standard errors and a negative binomial comparison.
5. **Check the binary flag.** Event study of weekly volume relative to normal days.
6. **Benchmark six forecasters** on three rolling-origin folds. Each fold trains only on data before its test window.
7. **Calibrate uncertainty** with 90% prediction intervals and coverage checks.
8. **Stress-test the conclusions:** bootstrap CIs on model differences, residual diagnostics, anomaly counts against chance, and segment-level tests.

Results

1. The peak is a sharp regime shift

Metric (daily mean)	Normal	Peak	Ratio
Transactions	32.0	88.0	2.75×
Revenue	17,914	72,172	4.03×
Units per order	2.02	3.49	1.73×
Average discount	7.5%	22.4%	2.98×
Average order value	559	819	1.46×

The event study shows a sharp step at the start of week 44 in all three years and no gradual ramp, so a binary peak flag is adequate.



Window scan (data before 2024-11-01 only). The assumed window ranks first, and the nearest alternative fits worse by 40.5 deviance points.

Window (start–end)	Deviance	Δ vs best
Nov 1 – Dec 30	637.5	0
Nov 1 – Dec 31	678.0	40.5
Nov 1 – Dec 25	896.0	258.5
Nov 5 – Dec 30	917.1	279.6

Model check. Pearson dispersion is 0.97 and the Cameron–Trivedi test finds no overdispersion ($p = 0.16$), so Poisson is appropriate. All three variants agree:

Model	Peak multiplier	95% CI
Poisson	2.75	2.70 – 2.80
Poisson (robust SE)	2.75	2.70 – 2.80
Negative binomial	2.75	2.70 – 2.80

2. Forecasting: the simple model holds up

Three folds, each trained only on earlier data: Peak 2024, Peak 2025 and a normal period (Jan–May 2026). Metric: daily revenue MAE (lower is better).

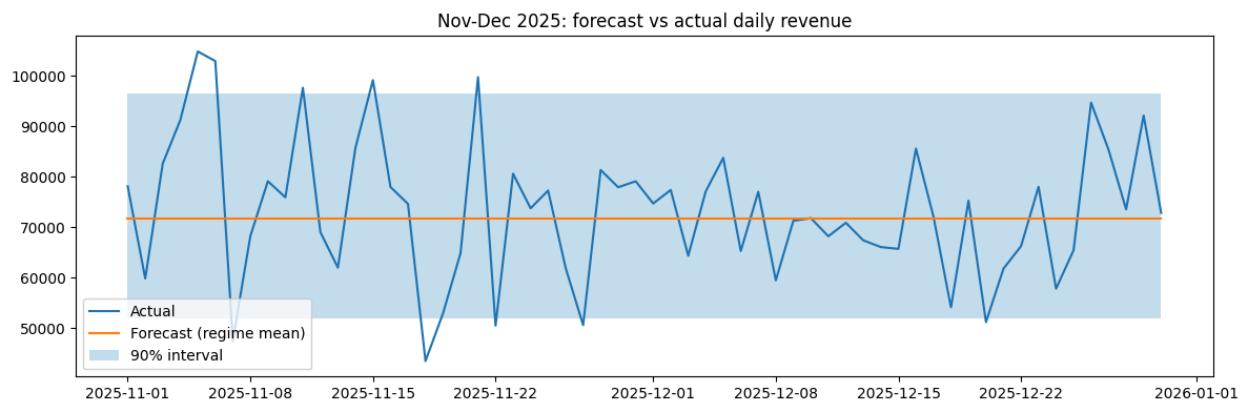
Model	Normal Jan–May 2026	Peak 2024	Peak 2025
Seasonal naive	6,684	16,914	15,848
Regime mean (peak/normal)	4,741	10,396	10,861
Regression (peak + trend + weekday)	4,848	10,550	10,901
SARIMAX(1,0,1) + peak flag	4,741	10,388	10,874

Model	Normal Jan–May 2026	Peak 2024	Peak 2025
LightGBM (calendar features)	5,308	10,234	11,904
Noise floor (oracle)	4,784	9,467	10,741

- The regime mean cuts error by **29%, 39% and 31%** versus seasonal naive in the three folds.
- It lands within about 1% of the oracle in the 2025 peak fold and within about 10% in the 2024 peak fold. Most of the remaining error is day-to-day noise that no model can remove.
- **Regime mean minus LightGBM**, bootstrap 95% CI on the MAE difference (negative means the regime mean is better):

Fold	MAE difference	95% CI	Reading
Peak 2024	+162	[-901, 1,187]	No detectable difference
Peak 2025	-1,043	[-2,233, 144]	No detectable difference
Normal 2026	-567	[-1,011, -121]	Regime mean better

MASE is above 1 in the peak folds (regime mean 1.59 and 1.52) but so is the oracle (1.45 and 1.50). That reflects high daily noise in peak relative to the seasonal-naive scale, not a weak model.



3. Prediction intervals

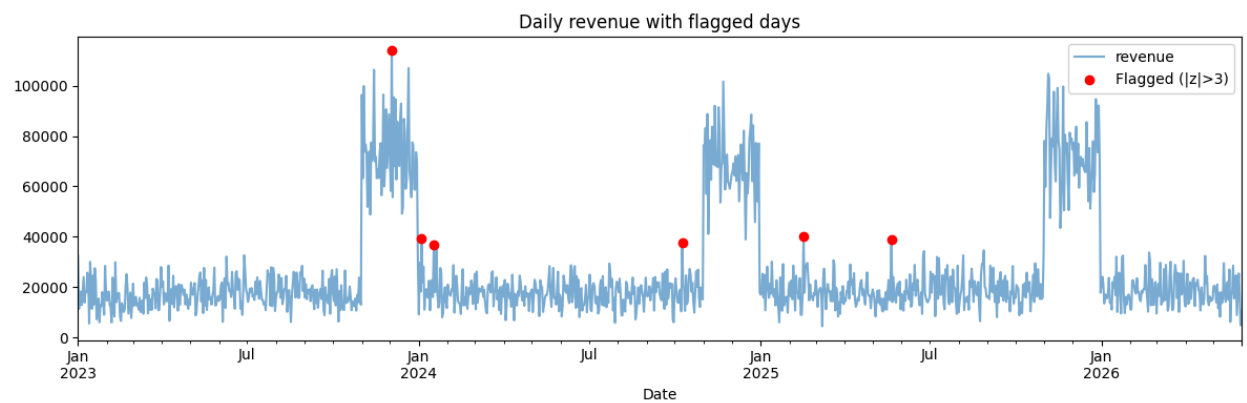
90% intervals built from the regime mean plus training-residual quantiles:

Fold	Coverage	Average width
Peak 2024	93.3%	48,352
Peak 2025	83.3%	44,566
Normal 2026	88.1%	18,673

Coverage is close to the 90% target. Peak 2025 is under by about 7 points, which is a mild miss given only 60 test days per fold.

4. Diagnostics and anomalies

- **Autocorrelation:** Ljung–Box p-values are 0.53 (lag 7), 0.03 (lag 14) and 0.22 (lag 28), so there is no consistent leftover structure.
- **Skew:** normal-period revenue is right-skewed (skew 0.52, Shapiro $p \approx 1e-9$).
- **Anomalies:** 6 days exceed $|z| > 3$ versus about 3.4 expected under normality. All six are upward spikes ($z = 3.0$ to 3.9) and none are drops. With right-skewed revenue, this is consistent with a heavy upper tail rather than evidence of true anomalies.



5. Planning numbers for the next peak season

	Revenue (60 days)
Past seasons	2023: 4.42M, 2024: 4.17M, 2025: 4.40M

	Revenue (60 days)
Baseline (mean daily revenue × 60)	4.33M
Bootstrap 90% interval (daily noise only)	4.16M – 4.51M
Year-to-year range	4.17M – 4.42M

The bootstrap interval and the observed year-to-year range agree. With only three past seasons, treat the interval as a planning range, not a guarantee.

6. Discount what-if (arithmetic, not a causal estimate)

Peak average discount is 22.4%. Within peak, discount and quantity are essentially uncorrelated ($r = 0.003$).

Scenario	Revenue per order	Break-even demand drop
22.4% → 15.0%	+9.6%	8.8%
22.4% → 7.5%	+19.3%	16.2%

Cutting the discount pays off only if peak demand falls by less than the break-even figure. The data cannot say how demand would respond, because peak days combine higher discounts with bigger baskets and no price experiment was run.

7. The lift is uniform across segments

Peak/normal ratios of daily order counts:

- **Location:** Germany 2.71, India 2.76, Pakistan 2.82, UAE 2.72, UK 2.72, USA 2.77
- **Product category:** Apparel 2.74, Beauty 2.77, Electronics 2.76, Home & Kitchen 2.73, Sports 2.76

No region or category drives the peak. Every segment rises by roughly the same factor. The tests use daily counts so each day counts once, with a Bonferroni correction.

Limitations

- **The data appears synthetic.** Sequential IDs, an exact revenue formula, near-Poisson daily counts, a uniform lift across every segment and a near-zero

discount–quantity correlation all point that way. The conclusions describe this dataset and may not transfer to real, messier data. In particular, the regime mean winning is expected when demand follows a simple regime structure.

- **Only two full peak seasons are available for forecast evaluation** (plus 2023 for training). The model comparison is therefore suggestive, which is why bootstrap intervals are reported instead of rankings alone.
 - **The window scan used a single training peak season (2023)**. The result matches the changepoint analysis but rests on limited data.
 - **No causal claims**. The discount analysis is break-even arithmetic only.
 - **The bootstrap treats days as independent**. The Ljung–Box results make this reasonable here, but it is still an approximation.
-

Repository contents

ecommerce_demand_forecasting.py	same code as a script
daily_transactions_and_revenue.png	daily transactions and revenue
nov_dec_2025_forecast_vs_actual_daily_revenue.png	forecast vs actual with 90% interval
daily_revenue_with_flagged_days.png	flagged days
weekly_transactions_relative_to_normal_day_mean.png	weekly volume relative to normal
results_mae.csv, results_mase.csv	forecast error tables
interval_coverage.csv	interval coverage by fold

How to run

The notebook was developed in Google Colab. Place the dataset at [/content/cleaned_global_ecommerce_data.csv](#), install `ruptures` with `pip install ruptures`, and run all cells. Other dependencies: `pandas`, `numpy`, `matplotlib`, `statsmodels`, `scipy`, `lightgbm`.

Tools

Python, pandas, statsmodels (GLM, SARIMAX, OLS, diagnostics), LightGBM, ruptures, SciPy, matplotlib.